

CMOS-compatible position-sensitive devices (PSDs) based on photodetector arrays

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Abstract

This paper reports five different constructions of optical position-sensitive devices (PSDs) implemented using standard CMOS technology. It is found that despite the non-idealities of CMOS-compatible photodetectors, CMOS technology provides a means of implementing PSDs with relatively high performance. This can be achieved, for example, by using an array of discrete photodetectors instead of the continuous single element structure used in conventional lateral effect PSDs (LEPs). The results show that, relative to a conventional LEP manufactured with dedicated technology, the linearity of an array-type two-axis CMOS PSD can be two to eight times better, and that its precision in low bandwidth (<10 kHz) applications can be 1.4 to 40 times better.

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1. Introduction

A position-sensitive device (PSD) is a photodetector capable of measuring the centroid position of a light spot falling on its surface. The same task can, of course, be performed by an imaging detector such as a CCD, but PSDs are typically favoured in high-volume applications where simple signal processing, high speed, robustness and low cost are of importance. Such applications include various triangulation-based distance sensors used in many industrial measurement tasks and consumer products such as compact cameras.

The most commonly used PSD is the lateral effect PSD (LEP). Despite dedicated and carefully optimised technology, the position sensing accuracy and sensitivity of LEPs are poorer than those of photodetector arrays such as CCDs. This is mainly due to the continuous single element structure of the conventional LEPs, whose implementation sets very high demands to the quality of the manufacturing technology in case high accuracy and sensitivity are desired. The purpose of this paper is to demonstrate, that the performance of LEPs can be improved by replacing the large single element photodetector with an array of discrete photodetectors and that such PSDs can be successfully implemented using standard CMOS technology.

The main goal of this work was to implement two-axis CMOS PSDs with good accuracy and incremental sensitivity. The devices were originally intended for sensor systems used outdoors where background illumination and the effect of atmospheric turbulence must be taken into account [5,6]. Modulated illumination of a few kHz is typically used in such cases to optimise noise performance and to prevent the background illumination from affecting measurement results. The effect of turbulence on measurement precision is reduced by using a small spot size on the detector.

Implementing PSDs using common CMOS technology instead of dedicated photodetector processes has two potential advantages: first, the compactness of the sensor can be improved by implementing all the electronics and the detector on a single chip and secondly, the sensor's overall performance may be improved using special PSD structures not realisable in conventional photodetector processes. The obvious drawback of CMOS technology is that it is not designed for light detection and thus does not provide best possible performance in that sense. However, our experiments show that despite the non-idealities of CMOS-compatible photodetectors, CMOS technology offers a possibility of implementing relatively high-performance PSDs.

This paper reports five different CMOS PSD constructions. These PSDs were implemented using standard 1.2 μm double metal, single poly-n-well CMOS technology. All the implementations described here are based on a current

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division principle similar to that of conventional LEPs, but in addition to the continuous structure, PSDs based on photodetector arrays are proposed. Previous work in the field has mostly concentrated on implementing CMOS LEPs based on the conventional single element structure [1–3], although Ref. [4] describes multi-pixel CMOS PSDs resembling those proposed here. The results presented in this paper will show that the performance of LEPs based on CMOS-compatible photodetector arrays can be markedly better than the performance of commercially available conventional LEPs.

2. Conventional LEP

LEPs provide an accurate and simple means for detecting the position of a light spot. The lateral position response of a one-axis LEP is inherently linear and its accuracy is not affected by spot truncation or non-optimal spatial sampling, as may be the case with devices composed of discrete photodetectors such as CCDs [7–9]. A one-axis LEP is basically a large-area photodiode with two extended electrodes at the opposite edges of a resistive layer acting as the anode or cathode of the device. Correspondingly, a tetralateral two-axis LEP has four extended electrodes at the edges of a single resistive sheet. A duolateral two-axis device has a resistive sheet on both terminals (anode and cathode) and a pair of electrodes on both sheets. The charge carriers generated in the illuminated region are divided between the electrodes in proportion to the conductance (distance) of the current paths between the illuminated region and the electrodes.

For a two-axis duolateral LEP, the deflection of the spot centroid from the centre of the device is simply:

$$\Delta x = \frac{L}{2} \frac{i_b - i_d}{i_b + i_d} \quad \text{and} \quad \Delta y = \frac{L}{2} \frac{i_a - i_c}{i_a + i_c} \quad (1)$$

where i_a , i_b , i_c , and i_d are the average signal currents of the electrodes a, b, c, and d, and L the length of the side of the LEP's active area (Fig. 1). The standard deviation of the position sensing results due to noise (i.e. the precision of a LEP) is

$$\sigma_{\text{LEP}} \approx \frac{L \sqrt{4kTB_N/R}}{P_S S} \quad (2)$$

where P_S is the RMS value of signal power, S the responsivity, B_N the noise equivalent bandwidth, and R the interelec-

trode resistance. The maximum value of R , which gives the lowest noise level for the PSD, is limited by the requirement that the sensor bandwidth $B_S = \pi/4RC$ (C equals device capacitance) must be clearly higher than the signal bandwidth to avoid linearity deterioration [10]. Linearity deterioration occurs since the bandwidth of a particular device terminal depends on spot position. Assuming that the noise bandwidth is $\pi/2$ times B_S , the limit for achievable precision in case of pulsed signal (signal band from dc to B_S) can be approximated using

$$\sigma_{\text{lim}} = \frac{L^2 B_S \sqrt{8kTC_{\text{sq}}}}{P_S S} \quad (3)$$

where C_{sq} is the LEP sheet capacitance.

According to Eq. (3), best precision is achieved by a photodetector with high responsivity, low sheet capacitance, and smallest active area suitable for the particular application. As dedicated photodetector processes provide high responsivity (up to 0.7 A/W) and low sheet capacitance (1–10 aF/ μm^2), reasonably high precision should be available if R is optimised. The problem is, however, that a relatively low sheet resistance value ($R \sim 10 \text{ k}\Omega/\square$) must be used to guarantee good resistance homogeneity and good linearity accordingly [11–13]. As a consequence, LEPs are noisy, and if good linearity is required, it is practically impossible to minimise noise by optimising interelectrode resistance, although the required detector bandwidth might allow it. Such limitations can be avoided in application specific integrated circuit (ASIC) design by using device structures that differ from that of conventional LEPs. The following chapters present five different PSD constructions suitable for CMOS ASIC designs.

3. CMOS LEP

CMOS photodetectors, such as well-substrate photodiodes, are suitable for optical sensing applications, where a responsivity range from 0.2 to 0.4 A/W is sufficient and a relatively large spatial and spectral non-uniformity in responsivity ($\sim 5\%$ standard deviation) can be tolerated, and where high speed and low crosstalk are not of importance [2,14–17]. The fairly large spatial responsivity variation of CMOS photodetectors (Fig. 2), which directly alters the electrical centroid position of the light spot, may be concluded to be detrimental to PSD linearity. However, experimental studies have shown that the linearity offered by CMOS PSDs utilising the LEP structure are equal to the best quality commercial LEPs (0.1–0.3% standard deviation with respect to full scale) [1,2,12,13]. Still, CMOS technology has serious drawbacks which affect the CMOS LEP performance. These include large junction capacitance of the available photodiode structures and low sheet resistance values of the layers suitable for current division. The former varies from 10 to 100 aF/ μm^2 and the latter is clearly less than $10 \text{ k}\Omega/\square$. The corresponding figures for LEPs manufac-

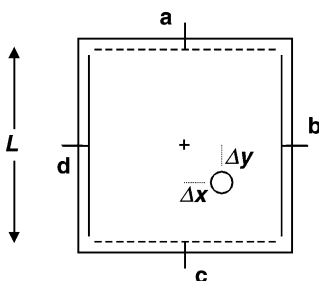


Fig. 1. Outline of a conventional two-axis PSD.

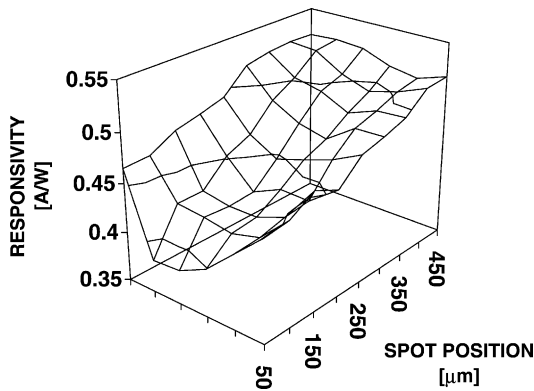


Fig. 2. Spatial responsivity variation of a CMOS photodiode.

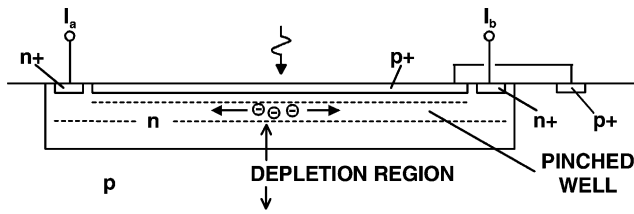
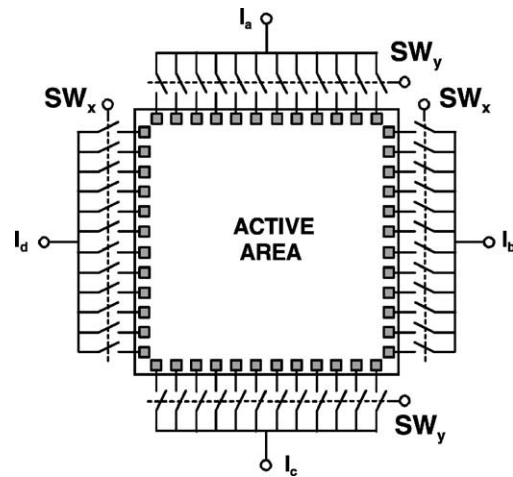


Fig. 3. Cross-section of the one-axis CMOS LEP.

tured with special processes are $1\text{--}10\text{ aF}/\mu\text{m}^2$ and $\sim 10\text{ k}\Omega/\square$ [12,13]. Large junction capacitance reduces the best achievable precision particularly in wide bandwidth ($>100\text{ kHz}$) applications, while low sheet resistance prohibits precision optimisation at lower bandwidths ($<10\text{ kHz}$).¹ In addition, standard CMOS makes it difficult to implement two-axis devices with good linearity. In the basic tetralateral structure, the current division between four contacts instead of two produces a large amount of non-linearity, while linear structures, such as duolateral [8] and pincushion [18], are practically impossible to implement in standard CMOS.

To demonstrate the performance of CMOS LEPs having conventional single element structure, this paper first presents the implementation and test results of a one- and two-axis CMOS LEP. The active area of the one-axis LEP measures $5.0\text{ mm} \times 0.2\text{ mm}$. It uses an n-well-p-substrate junction as a photodetector and a pinched well as the current dividing layer (Fig. 3). The measured responsivity of the well-substrate photodiode is 0.42 A/W at 850 nm and the interelectrode resistance is $152\text{ k}\Omega$ at the 5 V detector bias, which corresponds to the sheet resistance value offered by the pinched well resistors ($\sim 6\text{ k}\Omega/\square$). The noise equivalent power (NEP), defined as the optical power level that provides $\text{SNR} \sim 1$ simultaneously at each output terminal, is

¹ Due to the high sheet capacitance values of CMOS-compatible photodiodes, relatively low interelectrode resistance values must be used to maintain linearity in wideband applications, which makes CMOS LEPs noisier than equally sized conventional LEPs. In low bandwidth applications, optimal precision calls for high interelectrode resistance, which is typically not possible to reach due to the low sheet resistance values available.

Fig. 4. Simplified outline of the two-axis CMOS LEP. SW_x and SW_y are used to switch the opposite contact pairs on and off alternately.

$1.6\text{ pW}/\sqrt{\text{Hz}}$. The NEP is calculated on the basis of thermal noise and responsivity alone, since noise generated by dark current ($\sim 20\text{ pA}$) is negligible. The standard deviation of its integral non-linearity is about $8\text{ }\mu\text{m}$, when 80% of the active dimension is used for measurements (an error of 0.2% in the used scale). The linearity of the device is of the same order, but the NEP and the characteristic time constant of the CMOS LEP ($\tau = RC/\pi^2 \sim 2500\text{ ns}$, [10]) are larger than those of a conventional (non-CMOS) one-axis LEP of the same size (non-linearity $\sim 0.1\%$, NEP $\sim 1\text{ pW}/\sqrt{\text{Hz}}$, $\tau \sim 25\text{--}250\text{ ns}$, [12,13]).

The two-axis CMOS LEP was implemented using the same construction as the one-axis LEP. The inherent non-linearity of the tetralateral geometry was linearized by changing the two-axis PSD effectively to a one-axis device and by performing two-axis measurements using two successive one-axis measurements. The idea was originally demonstrated by Morikawa and Kawamura [19], but here we present a modified implementation suitable for CMOS technology. It utilises electrodes composed of multiple point contacts instead of continuous strip contacts. Each electrode is composed of a row of 100 minimum-sized n+ diffusion contacts positioned about $25\text{ }\mu\text{m}$ from each other. Each of the contact points is then connected to a common signal rail by a simple MOS switch. The operation is set-up by alternately disconnecting one pair of the electrodes. A simplified outline of the PSD measuring $2.5\text{ mm} \times 2.5\text{ mm}$ is depicted in Fig. 4.

Besides better linearity, the alternate operating principle and the use of multiple contact electrodes provide lower noise than the conventional tetralateral structure. This can be explained by the fact that the noise generating resistance at each preamplifier input increases when the virtual ground at the inputs of obsolete electrode pairs are disconnected and the low-resistive path formed by them breaks off.

The device's lateral transfer characteristics were measured using a $300\text{ }\mu\text{m}$ light spot with uniform intensity

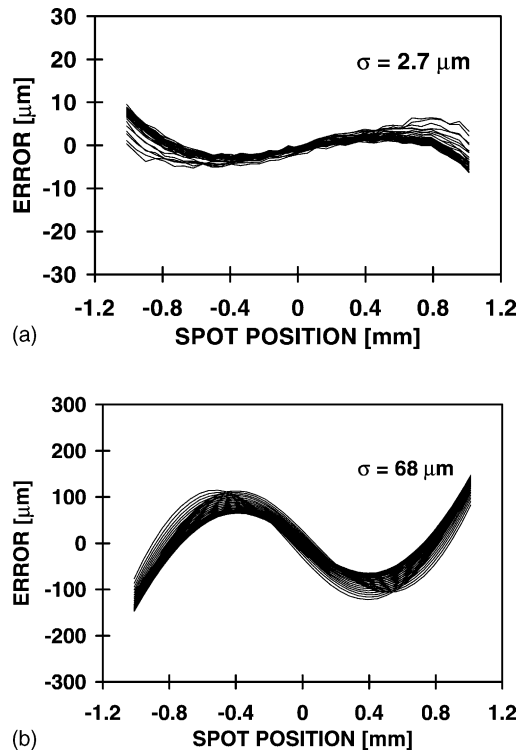


Fig. 5. Non-linearity error of the two-axis CMOS LEP along parallel scanning lines in (a) alternate (b) tetralateral mode.

distribution and 37×37 measurement points ($57 \mu\text{m}$ measurement grid) covering 80% of the PSD's active dimensions. As expected, non-linearity errors in the alternate ($2.7 \mu\text{m}$, 0.14%, Fig. 5(a)) and tetralateral ($68 \mu\text{m}$, 7.2%, Fig. 5(b)) modes corresponded to those of a conventional one-axis LEP and a conventional two-axis tetralateral LEP. In the alternate measurement mode, the interelectrode resistance and NEP were $4.5 \text{ k}\Omega$ and $10 \text{ pW}/\sqrt{\text{Hz}}$, respectively. In the tetralateral mode, on the other hand, the corresponding values were $0.85 \text{ k}\Omega$ and $47 \text{ pW}/\sqrt{\text{Hz}}$. In the alternate mode, linearity was better, whereas the NEP and the characteristic time constant (600 ns) were worse than those of a high-quality two-axis duolateral LEP (non-linearity $\sim 0.3\%$, $\text{NEP} \sim 5 \text{ pW}/\sqrt{\text{Hz}}$, $\tau \sim 6\text{--}60 \text{ ns}$, [12,13]). Note that the position sensitivity of the PSD could be somewhat improved by increasing interelectrode resistance using a mesh-type resistive layer for current division [20].

4. PSDs based on photodetector arrays

To demonstrate the possibility of implementing more sensitive PSDs in standard CMOS, we present three other devices comprising an area array of photodetectors and current dividing components residing outside the PSD's active area. This structure makes it possible to take advantage of resistors and photodetectors not suitable for the continuous single element LEP structure and therefore a means of im-

proving device position sensitivity, for example, by increasing interelectrode resistance or the effective responsivity of the photodetectors. First, however, the effect of discrete spatial sampling on the accuracy of centroid estimation is considered.

4.1. Effect of discrete spatial sampling on the accuracy of light spot centroid detection

Using an array of discrete photodetectors instead of the continuous structure can cause an additional centroid estimation error, particularly when the spot size is small. Alexander and Ng [7] have shown that the centroid of a spatially digitised light spot is free of systematic error only if the spatial bandwidth of that spot is less than the spatial sampling frequency of the array. Thus, problems usually arise when the size of the light spot approaches that of a single photodetector. Note, however, that crosstalk between the photodetectors decreases the spatial bandwidth of a light spot and thereby reduces the error caused by sampling.

An experiment was made to verify the existence of crosstalk. Fig. 6 shows the photocurrent distribution in a CMOS photodetector array composed of well-substrate photodiodes (pitch $25 \mu\text{m}$, fill factor 68%) when only one photodiode is illuminated (the rest are covered with metal). Note the wavelength dependence of crosstalk due to the different penetration depths of 633 and 850 nm light. The amount of crosstalk is particularly noticeable at 850 nm, where only half of the photogenerated carriers contribute to the current in the illuminated photodiode. The crosstalk is caused by the fact that the bottom of the depletion layer is relatively close to the surface, as a result of which a large amount of carriers reside within the substrate. Due to common substrate contact and the relatively long diffusion length ($>100 \mu\text{m}$) within the substrate, these carriers may freely diffuse to the neighbouring photodiodes.

4.2. Photodiode array PSD

The use of a photodiode array and a separate current dividing resistive network makes it possible to obtain higher interelectrode resistance and thus to achieve improved precision over the conventional LEP structure. The implemented photodiode array PSD consists of an area array of discrete well-substrate photodiodes and two linear arrays of MOS transistors residing outside the photosensing area. The idea of using MOS transistors for accurate current division in PSDs was originally proposed by Tartagni and Perona [21], who also demonstrated its performance. Their construction, however, did not include photodetectors. The method has also been applied to a position-sensitive array used to monitor solar illumination, but its performance in conjunction with the photodetector array was not fully characterised [22]. The current division technique employing MOS transistors provides inherently linear operation and the resistance of the current dividing MOS transistor array can be adjusted to

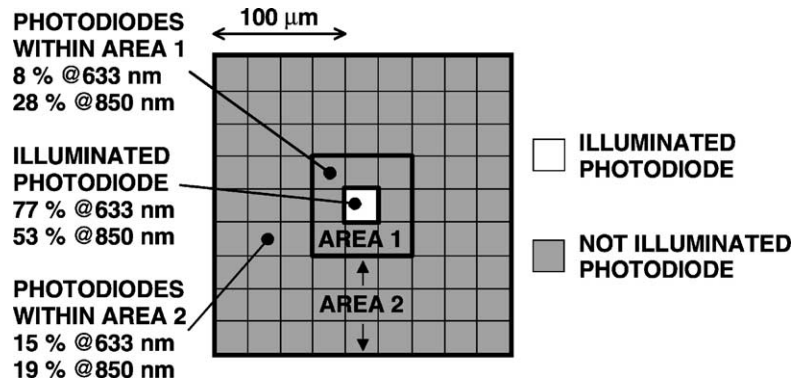


Fig. 6. Measured photocurrent distribution in a 9×9 well-substrate photodiode array when only one photodiode at the centre of the array is illuminated. The percentage of the photocurrent collected by the illuminated photodiode and the non-illuminated photodiodes surrounding it (AREA 1 and AREA 2) are given at the wavelengths of 633 and 850 nm.

any practical value simply by altering the transistor's W/L ratio [23]. The technique also provides a relatively compact method for implementing high resistances and enables the electrical adjustment of interelectrode resistance by changing the gate bias voltage of the transistors.

The PSD, whose construction is depicted in Fig. 7, is composed of 100×100 well-substrate photodiodes occupying a total area of $2.5 \text{ mm} \times 2.5 \text{ mm}$. The array pitch is $25 \mu\text{m}$ and the fill factor 68%. The cathode of the photodiodes in each row are alternately connected to the row and the column current lines. The row and column lines, again, are connected to two separate MOS transistor linear arrays, each consisting of 100 PMOS transistors used as resistors. In this prototype, the aspect ratio (100/20) of the transistors was designed to give an interelectrode resistance of a few hundred kilo-ohm and a bandwidth of about 10 kHz at 5 V gate bias. Relatively large-area transistors were used to obtain good linearity. The output currents of the PSD were processed like those of a conventional LEP.

The PSD's precision and linearity were measured with an LED source, which was square-wave modulated with

1.2 kHz and a narrow-band (30 Hz) synchronous receiver, whose noise spectral density was less than $0.1 \text{ pA}/\sqrt{\text{Hz}}$. The interelectrode resistance of the PSD was about $250 \text{ k}\Omega$ at the 5 V bias and about $800 \text{ k}\Omega$ at a 2 V bias. The noise level of the detector was $0.26 \text{ pA}/\sqrt{\text{Hz}}$ (5 V bias), which is about five times less than that of a conventional LEP ($R \sim 10 \text{ k}\Omega$). The NEP ($3.5 \text{ pW}/\sqrt{\text{Hz}}$) and the measured position sensing precision, however, were only 1.4 times better. The apparent discrepancy between the noise levels and the precision results is explained by the difference in current division and by the lower effective responsivity of the CMOS PSD. The signal current is divided into four parts in the array PSD instead of two as is the case in a conventional duolateral LEP. In addition, the responsivity of the PSD array, including the effect of its fill factor, was about 0.3 A/W , whereas the LEP used for comparison had a responsivity of 0.55 A/W at the 850 nm wavelength used in the measurements.

The result of the linearity measurement presented in Fig. 8 shows a standard deviation of $2.8 \mu\text{m}$ (0.14%), which is better than that measured for a high-quality commercial LEP ($6.3 \mu\text{m}$, 0.4%, Fig. 9). The symmetrical S-shape of the error curves suggests that the error results mostly from the fact that the receiver bandwidth is partially limited by

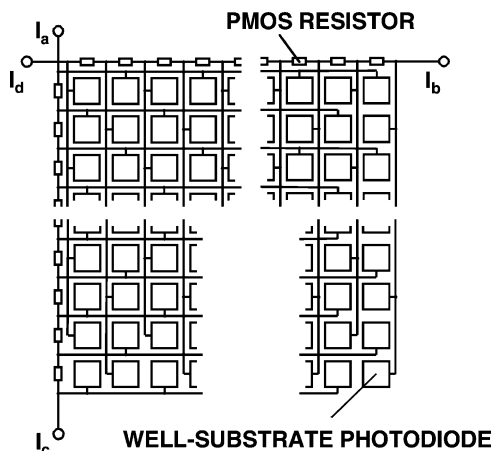


Fig. 7. Construction of the photodiode array PSD.

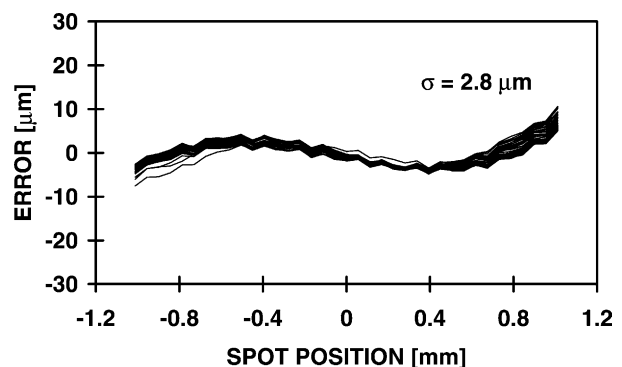


Fig. 8. Non-linearity error of the photodiode array PSD.

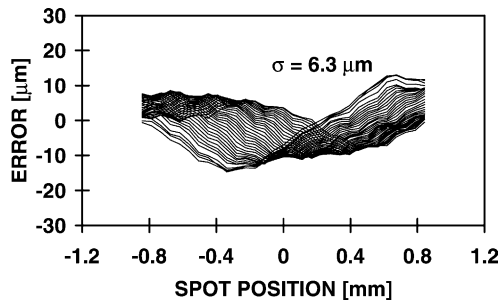


Fig. 9. Non-linearity error of a conventional two-axis LEP.

the detector. The MOS transistor mismatch would probably have produced non-symmetrical error curves, and the spatial non-uniformity of photodiode responsivity would have resulted in curves with a clearly different shape in each parallel scanning line. Hence, it seems that non-uniform responsivity and MOS transistor mismatch have only a minor effect on accuracy in this case. Note that due to the similarity of the lateral responses, a simple error correction table would be enough to improve the PSD's accuracy, which is not the case with a conventional LEP whose lateral responses along each scanning line clearly differ from each other as depicted in Fig. 9.

The highest usable signal frequency was roughly estimated by measuring the lateral transfer characteristics of the device also at 4.8 and 9.6 kHz modulation frequency. The results indicate that the highest possible signal frequency at which the measured accuracy can be obtained is about 5 kHz when the interelectrode resistance is 250 kΩ (5 V bias). This result agrees with simulations and the measured bandwidth (12 kHz). The bandwidth of the PSD could be increased by decreasing the area of the current dividing transistors whose contribution to total capacitance (400–500 pF) is substantial (~300 pF).

To find out additional errors that might be caused by the discrete photodetector structure, a set of measurements were performed using 50 and 300 μm light spots and a high measurement point density (5.7 μm). The results, depicted in Fig. 10a (50 μm) and Fig. 10b (300 μm) show that the 50 μm spot clearly suffers from a periodical error whose period length corresponds to that of the array pitch. As for the larger spot such error is indistinguishable from other non-linearities. According to the simulation results presented in Ref. [24], the smallest possible centroiding error with the 50 μm spot should have been larger than 1/10 of the array pitch. The low measured error value (0.7 μm, 1/36 of the array pitch) is probably the result of crosstalk as explained before. It can be concluded that, in position sensing applications in which the spot size typically varies from 0.1 to 1 mm, the use of a CMOS photodetector array instead of the continuous structure does not seriously increase the measurement error. As a consequence, the array structure seems fully suitable for implementing accurate PSDs.

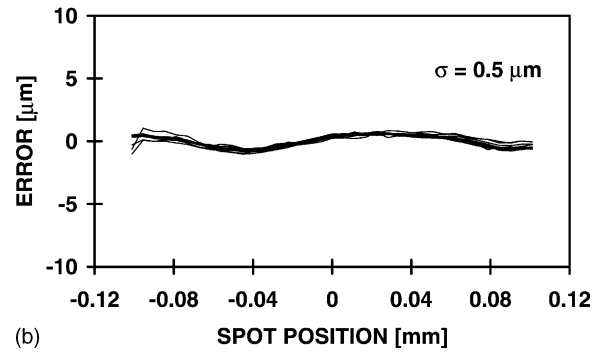
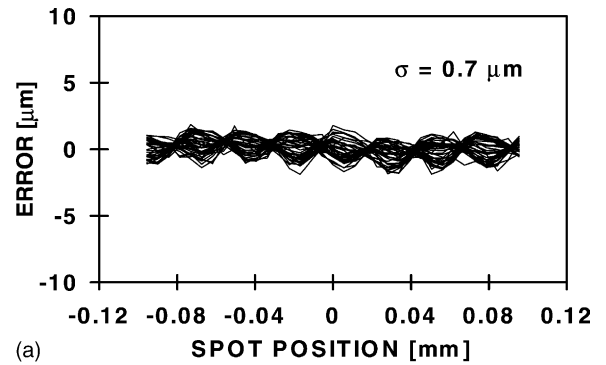


Fig. 10. Error caused by discrete photodetector structure when the spot diameter is (a) 50 μm and (b) 300 μm.

4.3. Phototransistor PSD

According to Eq. (2) another practical way of improving precision is to increase the responsivity of the photodetector array. In CMOS, one option is to use phototransistors, in which case the precision for a two-axis device is

$$\sigma \approx \frac{2L\sqrt{4kTB_N/R + 2q(P_S + P_B)S\beta^2 B_N}}{P_S S\beta} \quad (4)$$

where β is the small signal current gain of the phototransistors, S their primary responsivity, P_B the background illumination power, and R the interelectrode resistance. It can be concluded that even a moderate value of β effectively improves precision over a conventional LEP, provided that the background illumination is relatively low.

The other advantage of the phototransistor PSD is the possibility to use low interelectrode resistance and thereby increase the bandwidth of the current dividing network to avoid the non-linearity produced by its finite bandwidth. Note that the increased level of noise due to lower interelectrode resistance is readily compensated for by the gain of the phototransistors which amplify the signal before the noisy current division.

The overall construction of the phototransistor PSD illustrated in Fig. 11 is similar to that of the photodiode PSD. It is composed of 100×100 CMOS-compatible vertical phototransistors (pitch 25 μm and fill factor 68%) occupying a total area of 2.5 mm $\times$ 2.5 mm. The emitters of the transistors are connected to the row and column current lines, which in

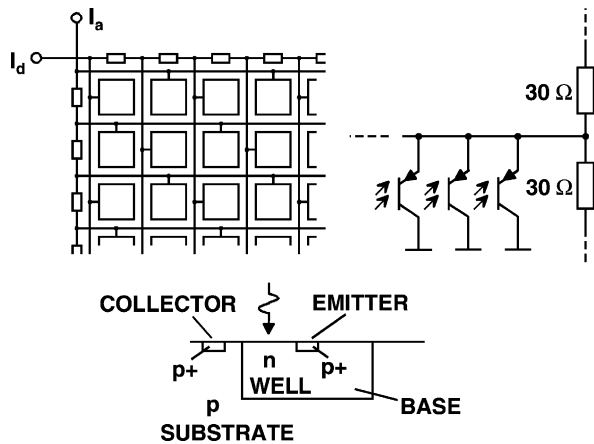


Fig. 11. Construction of the phototransistor PSD.

turn are connected to two arrays of resistors composed of a 100 $30\ \Omega$ polysilicon resistors each. Polysilicon resistors were used since they provide lower capacitance and also a more compact design than MOS resistors.

Precision was tested using low background illumination, a spot size of $50\ \mu\text{m}$ and a light source modulation frequency of $1.2\ \text{kHz}$. When compared experimentally to a duolateral LEP ($R \sim 10\ \text{k}\Omega$), it was found that the responsivity of the phototransistor PSD is high enough to provide approximately seven times better precision at best ($P_S \sim 10\ \text{nW}$) and that equal precision ($\sigma/L \sim 1\%$) can be achieved using ten times less signal power at best (Fig. 12). Effective responsivity ($S\beta$) can still be increased at low signal levels using either background or a signal to optically bias the phototransistors (β increases as the dc component of the collector current is raised from zero). Due to a smaller illuminated area, the latter method results in a lower total bias current and consequently, in better SNR and precision. By using a $100\ \text{pW}$ signal bias responsivity in excess of $10\ \text{A/W}$ was achieved even at very low signal levels ($< 10\ \text{pW}$). The NEP without bias was about $0.5\ \text{pW}/\sqrt{\text{Hz}}$. The standard deviation of the non-linearity error was $0.7\ \mu\text{m}$ (0.04% , Fig. 13) and $1.1\ \mu\text{m}$ when using 300 and $50\ \mu\text{m}$ light spots, respectively. Note that the error is clearly smaller than that of

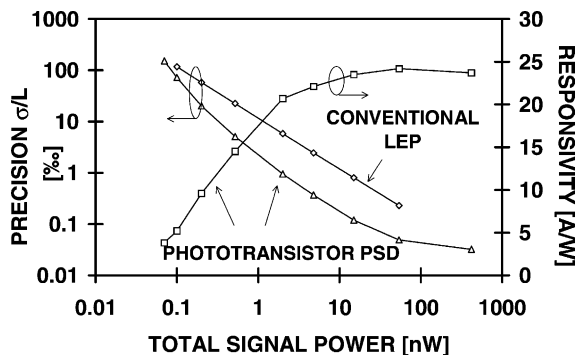
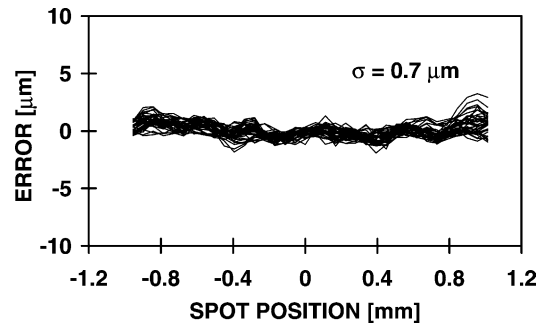


Fig. 12. Comparison of the position sensing precision between the phototransistor PSD and a conventional LEP.

Fig. 13. Non-linearity error of the phototransistor PSD using $300\ \mu\text{m}$ spot diameter.

the photodiode array PSD and that the error curve has no S-shape, which suggests that the non-linearity due to finite PSD bandwidth is negligible. The shape of the error curves implies that the spatial non-uniformity of responsivity and the resistor mismatch both contribute to it.

4.4. Tracking PSD

The main idea of the tracking PSD was to combine the best properties of two common PSD types: quadrant detector (QD) PSDs' high electrical precision and LEPs' linear measurement field and low sensitivity to atmospheric turbulence and background noise. The properties of LEPs result from two factors: the possibility of using a focused rather than a defocused light spot and the smaller active dimensions necessary to cover a particular measurement field [5,6]. A QD consists of four photodiodes (quadrants) positioned symmetrically around the centre of the device and separated by a narrow gap. Position information is derived from the optical signal power received by the quadrants, whose electrical contribution then serves to define the relative position of the light spot with respect to the gap position. The operation of the proposed tracking PSD is similar to that of the QD, with the exception that the gap electrically tracks the light spot within the active area of the PSD (Fig. 14).

The high accuracy ($\ll \mu\text{m}$) of the photodetector array geometry is used to establish good integral linearity, while high precision is based on "subspot" interpolation based on the

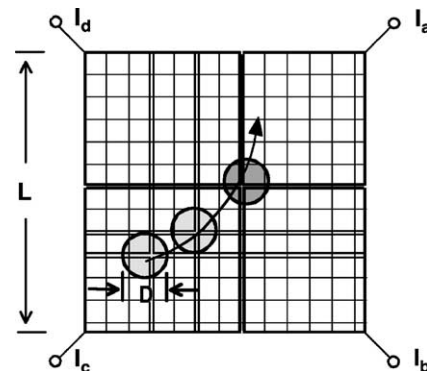


Fig. 14. Operating principle of the tracking PSD.

inherently non-linear but very sensitive QD principle. The approximations of the displacements with respect to the gap position near the measurement field centre are

$$\Delta x \approx \frac{\pi D}{8} \frac{(i_a + i_b) - (i_c + i_d)}{i_a + i_b + i_c + i_d} \quad \text{and} \quad \Delta y \approx \frac{\pi D}{8} \frac{(i_a + i_d) - (i_c + i_b)}{i_a + i_b + i_c + i_d} \quad (5)$$

where i_a , i_b , i_c and i_d are the average signal currents of the quadrants a, b, c and d, and D the diameter of a uniform round spot. The standard deviation of the measured spot positions due to noise is

$$\sigma \approx \frac{\pi D i_n}{4 P_S S} \quad (6)$$

where i_n is typically caused by the feedback resistance of the transimpedance preamplifier ($>1 \text{ M}\Omega$) or by background illumination. Note that a LEP's scale factor is solely determined by the extent of the device's active dimensions L , whereas in a tracking PSD, the scale factor of the interpolation function is determined by the size of the light spot D , which is typically only a small fraction of the device's active dimensions. Thus, assuming equal responsivity and active area size, the precision of a tracking PSD compared to that of a LEP (σ_{LEP}) is

$$\sigma \approx \frac{\pi}{4} \frac{D}{L} \frac{i_n}{i_{n\text{LEP}}} \sigma_{\text{LEP}} \quad (7)$$

where i_n and $i_{n\text{LEP}}$ are the noise of the tracking PSD and the LEP, respectively. Since D/L and $i_n/i_{n\text{LEP}}$ are typically of the order of 0.1, the precision of the tracking PSD could well be two orders of magnitude higher than that of an LEP.

The construction of the tracking PSD depicted in Fig. 15 is essentially similar to that of the photodiode array PSD. This time, the transistors are used to divide the array into four separate areas. In the prototype, the transistor gates are driven by the logic signals of a shift register. As the output of each shift register drives four PMOS transistors, the smallest possible gap width and gap step are $100 \mu\text{m}$.

A spot size of $300 \mu\text{m}$ was used during the tests to properly cover the minimum gap step. The non-linearity of the

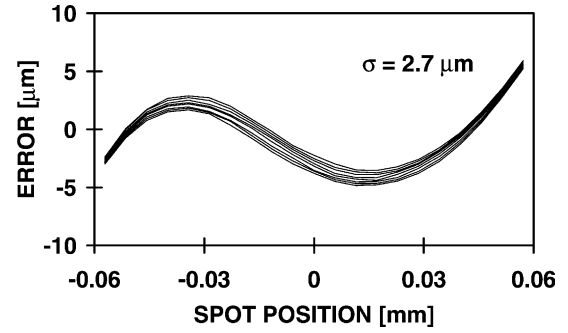


Fig. 16. Error of "subspot" interpolation of the tracking PSD.

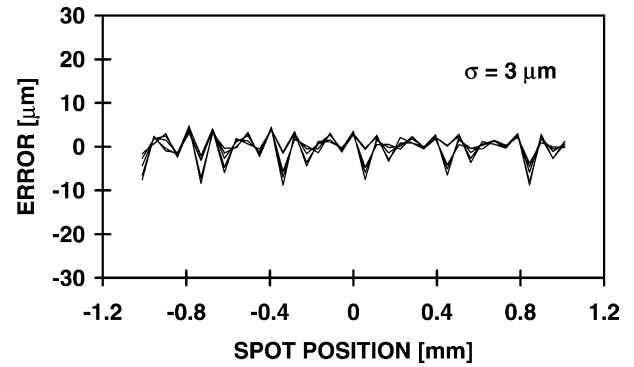


Fig. 17. Position sensing error of the tracking PSD.

interpolator was $2.7 \mu\text{m}$ (standard deviation) when approximately half ($\sim 100 \mu\text{m}$) of its effective measurement span was used (Fig. 16). As illustrated by Fig. 17, the PSD shows excellent linearity. The non-linearity of the interpolation causes a ripple, but with a standard deviation of only $3 \mu\text{m}$; thus, the PSD offers better accuracy than an equally sized conventional LEP, whose non-linearity range varies from 5 to $10 \mu\text{m}$ at best [12,13]. Note that the absolute value of the error does not increase as the device size L (measurement field) increases, and that the error can be further reduced by using a smaller gap width and a smaller spot diameter. The measured NEP was $1.3 \text{ pW}/\sqrt{\text{Hz}}$ and the corresponding position sensing precision about 40 times better than that of a conventional duolateral LEP of the same size. Owing to the

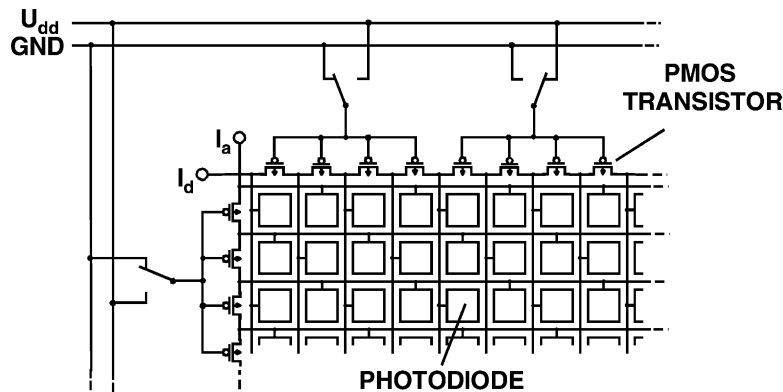


Fig. 15. Construction of the tracking PSD.

Table 1
Properties and performance of conventional LEPs and the implemented CMOS PSDs^a

Property		Conventional LEP	One-axis CMOS LEP	Two-axis CMOS LEP	Photodiode PSD	Phototransistor PSD	Tracking PSD
Responsivity S (A/W)		0.5–0.7	0.42	0.38	0.3 ^b	>20 ^c	0.3 ^b
Sheet resistance ($k\Omega/\square$)		10	6	6			
Interelectrode Resistance R ($k\Omega$)	One-axis	250	152	4.5 ^d			
	Two-axis	10		0.85 ^e	250	3	
NEP ($\text{pW}/\sqrt{\text{Hz}}$)	One-axis	~1	1.6	10 ^d			
	Two-axis	~5		47 ^e	3.5	0.5	1.3
Capacitance C (pF)	One-axis	1–10	160				
	Two-axis	6–60		1300	~400		
Time constant RC/π^2 (ns)	One-axis	25–250	2500	600 ^d			
	Two-axis	6–60		110 ^e	~10000	~100000	
Non-linearity (%)	One-axis	~0.1	0.2	0.14 ^d			
$FS = 0.8 \times L$	Two-axis	~0.3		7.2 ^e	0.14	0.06 ^c	0.15

^a The figures for the conventional LEPs are deduced from manufacturer data sheets assuming that one- and two-axis LEPs would measure $5 \text{ mm} \times 0.2 \text{ mm}$ and $2.5 \text{ mm} \times 2.5 \text{ mm}$, respectively. The test results of the CMOS PSDs were obtained using a low-noise receiver, a modulation frequency of 1.2 kHz, a wavelength of 850 nm and a spot size of 300 μm .

^b Includes fill factor.

^c $P_S > 1 \text{ nW}$, spot diameter 50 μm .

^d Alternate mode.

^e Tetralateral mode.

lower effective responsivity of the CMOS PSD, the difference is smaller than predicted by Eq. (7). Note again that the achievable precision is not dependent on the extent of the measurement field, therefore the PSD achieves much higher precision-to-measurement field ratios (σ/L) than an LEP.

5. Conclusions

Five different CMOS-compatible PSDs utilising either a continuous or a discrete array structure were implemented and characterised. The properties of CMOS technology are mainly dictated by the requirements of digital electronics and thus it is obviously not an ideal technology for the implementation of photodetectors. The non-idealities of its photosensing properties include, for example, low responsivity, relatively high spatial responsivity variation, and large photodetector capacitance. The results summarised in Table 1 shows, however, that despite these non-idealities, CMOS technology provides a means for implementing accurate and sensitive PSDs. CMOS ASIC technology allows the construction of new types of PSDs without the limitations posed by the conventional LEP structure. Constructions based on an array of photodetectors, for example, make it possible to separate current division and photodetection, and to achieve better performance by using resistors and photodetectors other than those suitable for the continuous LEP structure. The array PSDs were shown to provide as much as 40 times better precision in low bandwidth (<10 kHz) applications and up to eight times better linearity than a conventional commercially available LEP. The properties of CMOS technology which make the above results

possible includes small feature size, accurate geometry, uniform electrical characteristics, and the possibility of embedding active devices within the PSDs. Also the high level of crosstalk between photodetectors, which generally has a detrimental effect on the performance of optical detector arrays, improves the centroiding accuracy of an array PSD by reducing the error caused by discrete spatial sampling.

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